THE ADDITIVE LADDER: CONGRUA SETS, GAUSSIAN INTEGERS, AND PARTIAL IMPOSSIBILITY THEOREMS FOR 3×3 MAGIC SQUARES OF SQUARES

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Abstract. Whether a 3×3 magic square with nine distinct square entries exists is a well-known open problem. Any such square has center m^2 and row/column offsets lying in the *congrua set* $D(m) = \{2ef : e^2 + f^2 = m^2\}$, and its corners are square if and only if $D(m)$ contains an additive quadruple $U, V, U+V, U-V$. We prove that $D(m)$ admits *no additive relation whatsoever* — no solution of $\varepsilon_1 d_1 + \varepsilon_2 d_2 + \varepsilon_3 d_3 = 0$ — whenever the split part of m (its largest divisor supported on primes $\equiv 1 \bmod 4$) is p^a, pq, or p^2q. Consequently *the center of any 3×3 magic square of squares is divisible by three distinct primes $\equiv 1 \bmod 4$, or its split part is of the form p^3q or p^2q^2.* The proofs pass through a reformulation of additive relations as six-term vanishing sums on a norm torus in $\mathbb{Z}[i]$, and are ultimately carried by elementary machinery: valuations in free multiplicative groups, an exact tangent-half-angle factorization calculus, divisibility trees, and the classical quartic descents of Fermat — both $x^4 - y^4 = z^2$ and $x^4 + y^4 = z^2$ occur as terminal cases, alongside the non-congruence of 2 and 3. Every enumeration, identity, and case in the proof is machine-verified; the verification suite and all artifacts are public.

1. Introduction

A *magic square of squares* (MSS3) is a 3×3 magic square whose nine entries are distinct perfect squares. Whether one exists is open. Classical analysis reduces the problem to arithmetic of *congrua*: the entries are $m^2 \pm U$, $m^2 \pm V$, $m^2 \pm U \pm V$ and m^2, and the row/column conditions force $U, V \in D(m)$ where

$$D(m) \;=\; \{\, 2ef \;:\; e^2 + f^2 = m^2,\; e, f > 0 \,\}.$$

The corner entries are squares if and only if additionally $U + V \in D(m)$ and $U - V \in D(m)$: an *additive quadruple* inside a single congrua set. In particular an MSS3 yields an *additive triple* $d_1 + d_2 = d_3$ in $D(m)$, so the following conjecture implies that no MSS3 exists.

Conjecture (A3.C). No congrua set $D(m)$ contains an additive triple.

Exhaustive computation verifies A3.C for all $m \le 10^7$. This paper proves it unconditionally on three infinite families, organized by the *split part* of m: writing $m = 2^s r \prod p_i^{a_i}$ with r supported on primes $\equiv 3 \bmod 4$ and $p_i \equiv 1 \bmod 4$ distinct, the split part is $\prod p_i^{a_i}$. Only the split part matters: $D(m)$ is a scalar multiple of the congrua set of the split part.

Theorem 1.1 ($\omega = 1$)**.** *If the split part of m is a prime power p^a, then no relation $\varepsilon_1 d_1 + \varepsilon_2 d_2 + \varepsilon_3 d_3 = 0$ holds with $d_i \in D(m)$, $\varepsilon_i \in \{\pm 1\}$ (repetitions allowed).*

Theorem 1.2 (pq)**.** *The same holds if the split part is pq with $p \neq q$.*

Theorem 1.3 (p^2q)**.** *The same holds if the split part is p^2q.*

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Corollary 1.4. *The center of any* 3×3 *magic square of squares is divisible by three distinct primes* $\equiv 1 \bmod 4$*, or has split part of the form* p^3q *or* p^2q^2 *with* $p,q\equiv 1 \bmod 4$.

The proofs share a five-layer architecture which appears to close every exponent box (a,b) it has been applied to, with residual case families visibly parametric in (a,b); this suggests a uniform theorem for all m with at most two distinct split primes (§**??**).

Method. Additive relations in $D(m)$ are exactly six-term vanishing sums $\sum_j \varepsilon_j(w_j - m^4/w_j) = 0$ with $w_j = z_j^2$ on the norm-m^4 torus of $\mathbb{Z}[i]$ (§**??**). For split part p^aq^b the w_j are monomials $\sigma^j\tau^k$ in a free rank-two subgroup of $\mathbb{Q}(i)^\times$, and a relation is a vanishing sum of at most six monomials. The layers: (1) *valuation pruning* in the free group kills most sign patterns outright; (2) an exact *tangent-half-angle calculus* — $\sigma = (1+it_1)/(1-it_1)$ with t_1 *rational* — converts each surviving pattern into an integer polynomial identity which, for coherent patterns, factors completely into terms that would force $\sigma^\alpha\tau^\beta = \pm 1$, impossible in a free group; (3) finite *congruence sieves* under Pythagorean side conditions; (4) *Gaussian collapse*: identities such as $p^{2d} \pm \ell^{2d} = \ell^d(\bar\ell^d \pm \ell^d)$ ($\ell = \lambda^2$, λ the Gaussian prime over p) compress the remaining relations to statements a q-adic valuation kills; (5) *divisibility trees* whose terminal cases are classical: the two Fermat quartic theorems, the non-congruence of 2 and 3, and short explicit descents.

Verification. Every enumeration and every identity in this paper is machine-verified in an open suite (148 checks at the time of writing); §**??** describes the methodology, including an enumeration-completeness audit performed after a gap — omission of the all-equal sign class — was caught by the suite's own consistency checks and repaired. We regard enumeration completeness as a theorem obligation, and prove it by comparison against an enumeration-independent brute construction.

2. Setup and the $\mathbb{Z}[i]$ reformulation

With $z = e + if$, $\mathrm{Im}(z^2) = 2ef$, so $D(m) = \{\,|\mathrm{Im}(z^2)| : z\in\mathbb{Z}[i],\ |z|^2 = m^2\,\}$. Writing $w_j = z_j^2$, so that $|w_j| = m^2$ and $\bar w_j = m^4/w_j$, a signed additive relation is exactly the six-term vanishing sum $\sum_j \varepsilon_j(w_j - m^4/w_j) = 0$. For split part p^aq^b, unique factorization in $\mathbb{Z}[i]$ gives

$$D(m) = \left\{\, m^2\left|\mathrm{Im}(\sigma^j\tau^k)\right| \;:\; (j,k)\in[-a,a]\times[-b,b]\setminus\{0\}\,\right\}, \qquad \sigma = \frac{\lambda^4}{p^2}, \quad \tau = \frac{\mu^4}{q^2},$$

where λ,μ are Gaussian primes over p,q. The valuations $v_{\lambda,\bar\lambda,\mu,\bar\mu}(\sigma^j\tau^k) = (2j,-2j,2k,-2k)$ show $\langle\sigma,\tau\rangle$ is free of rank two and no monomial is ±1.

Lemma 2.1 (equal-modulus rigidity)**.** *No three nonzero elements of* $\mathbb{Q}(i)$ *of equal absolute value satisfy* $\pm a \pm b \pm c = 0$.

Proof. Absorbing signs, $a + b = c$; dividing by c gives $\alpha+\beta = 1$ with $|\alpha| = |\beta| = 1$ in $\mathbb{Q}(i)$, whence $\mathrm{Re}\,\alpha = \frac{1}{2}$ and $\alpha = \zeta_6^{\pm1}\notin\mathbb{Q}(i)$. □

Proposition 2.2 (degenerate subsums)**.** *Every vanishing proper subsum of the six-term sum has size* 2 *or* 4*, and a size-*2 *vanishing forces equal congrua. Hence genuine additive triples of distinct congrua give nondegenerate vanishing sums.*

Proof. Sizes $1,5$ are impossible (all terms have modulus m^2), size 3 by Lemma **??**; a size-2 relation forces $w'\in\{\pm w, \pm\bar w\}$, giving $|\,\mathrm{Im}\,w'| = |\,\mathrm{Im}\,w|$. □

3. One split prime: proof of Theorem ??

With a single split prime, $D(m) = \{\, m^2|\operatorname{Im}\sigma^k| : 1 \le k \le a \,\}$, where $\sigma = \lambda^4/p^2$ satisfies the primitive integer polynomial $R(x) = p^2x^2 - 2Cx + p^2$ with $C = \operatorname{Re}\lambda^4$ and $\gcd(C,p) = 1$ (if $\lambda \mid 2C = \lambda^4 + \bar{\lambda}^4$ then $\lambda \mid \bar{\lambda}^4$, contradicting coprimality). A signed relation becomes, after absorbing orientations, $\sum_i \varepsilon_i'(\sigma^{k_i} - \sigma^{-k_i}) = 0$. If all k_i are equal this reads (odd integer) $\cdot\, 2i\operatorname{Im}\sigma^K = 0$ with $\operatorname{Im}\sigma^K \neq 0$: impossible. Otherwise multiplying by σ^K, $K = \max k_i$, exhibits σ as a root of

$$Q(x) = \sum_i \varepsilon_i'\big(x^{K+k_i} - x^{K-k_i}\big),$$

an integer polynomial with coefficients in $\{0, \pm1, \pm2, \pm3\}$ and $\operatorname{lc}(Q) = -Q(0) \neq 0$. Since σ is nonreal, its minimal polynomial over $\mathbb{Q}$ is $x^2 - (2C/p^2)x + 1$, with primitive integer form R. By Gauss's lemma $Q = \gamma R T$ with $T \in \mathbb{Z}[x]$ and $\gamma = \operatorname{cont}(Q) \le 3$; comparing leading coefficients, $|\operatorname{lc}(Q)| \le 3 < p^2 \le |\gamma\, p^2 \operatorname{lc}(T)|$ — impossible for $p \ge 5$. □

4. Two split primes I: the framework and the (1, 1) box

5. Two split primes II: the (2, 1) box

6. Machine verification and the completeness audit

7. Outlook

The residual case families of every box analyzed so far are parametric in the exponent box, and the five-layer machinery is uniform; we conjecture and are pursuing a single theorem for all m with at most two distinct split primes. The known-m frontier of the MSS3 problem itself now lies at three or more distinct split primes, where the monomial group has rank three; whether the present methods extend is the natural next question.